Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030597

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Query 1: - CREATE DATABASE

```
2520030597-Vidhath

CREATE DATABASE IF NOT EXISTS joins_db;
USE joins_db;
```

Output: -

#	Time	Action	Message	Duration / Fetch
1	13:55:57	CREATE DATABASE IF NOT EXISTS joins_db	1 row(s) affected	0.031 sec
2	13:56:01	USE joins_db	0 row(s) affected	0.000 sec

Query 2: -CREATE TABLES

```
CREATE TABLE class (
    id INT,
    name VARCHAR(30)
);
CREATE TABLE class_info (
    id INT,
    address VARCHAR(30)
);
```

Output: -

3	13:58:04	CREATE TABLE class (id INT, name VARCHA...	0 row(s) affected	0.125 sec
4	13:58:07	CREATE TABLE class_info (id INT, address V...	0 row(s) affected	0.047 sec

Query 3: -INSERT VALUES

```
2520030597-Vidhath
INSERT INTO class VALUES
(1,'Vidhath'),
(2,'Raghuveer'),
(4,'Amulya');

INSERT INTO class_info VALUES
(1,'HYDERABAD'),
(2,'DELHI'),
(3,'CHENNAI');
```

Output: -

7	14:01:03	INSERT INTO class VALUES (1,'Aswitha'), (2,'Adhr...	3 row(s) affected Records: 3 Duplicates: 0 Warnin...	0.031 sec
8	14:01:08	INSERT INTO class_info VALUES (1,'HYDERABA...	3 row(s) affected Records: 3 Duplicates: 0 Warnin...	0.062 sec

Query 4: - CROSS JOIN

```
2520030597-Vidhath
SELECT *
FROM class
CROSS JOIN class_info;
```

Output: -

17	14:04:24	SELECT * FROM class CROSS JOIN class_info L...	9 row(s) returned	0.000 sec / 0.000 sec

Result Grid | Filter Rows: | Export:

id	name	id	address
4	Amulya	1	HYDERABAD
2	Raghuveer		HYDERABAD
1	Vidhath		HYDERABAD
4	Amulya	2	DELHI
2	Raghuveer		DELHI
1	Vidhath		DELHI
4	Amulya	3	CHENNAI
2	Raghuveer		CHENNAI
1	Vidhath		CHENNAI

Result 1

Query 5: - INNER JOIN

```sql
SELECT *
FROM class
INNER JOIN class_info
ON class.id = class_info.id;
```

Output: -

18 14:08:29 SELECT * FROM class INNER JOIN class_info O... 2 row(s) returned 0.000 sec / 0.000 sec

id	name	id	address
1	Vidhath		HYDERABAD
2	Raghuveer		DELHI

Query 6: -INNER JOIN

```sql
2520030597-Vidhath
SELECT class.name,
       class_info.address
FROM class
INNER JOIN class_info
ON class.id=class_info.id;
```

Output: -

19 14:10:36 SELECT class.name, class_info.address FRO... 2 row(s) returned 0.000 sec / 0.000 sec

Result Grid | Filter Rows:

name	address
Vidhath	HYDERABAD
Raghuveer	DELHI

Query 7: - NATURAL JOIN

```sql
SELECT *
FROM class
NATURAL JOIN class_info;
```

Output: -

20	14:12:24	SELECT * FROM class NATURAL JOIN class_inf...	2 row(s) returned	0.000 sec / 0.000 sec

id	name	address
1	Vidhath	HYDERABAD
2	Raghuveer	DELHI

Query 8: - LEFT OUTER JOIN

```
INSERT INTO class VALUES
(5,'Nihitha');
INSERT INTO class_info VALUES
(7,'BENGALURU'),
(8,'PUNE');
```

Output: -

21	14:14:30	INSERT INTO class VALUES (5,'Nihitha')	1 row(s) affected	0.031 sec
22	14:14:36	INSERT INTO class_info VALUES (7,'BENGALU...	2 row(s) affected Records: 2 Duplicates: 0 Wami...	0.046 sec

Query 9: - LEFT JOIN

```
SELECT *
FROM class
LEFT OUTER JOIN class_info
ON class.id=class_info.id;
```

Output: -

23	14:16:23	SELECT * FROM class LEFT OUTER JOIN class...	4 row(s) returned	0.000 sec / 0.000 sec

id	name	id	address
1	Vidhath		HYDERABAD
2	Raghuveer		DELHI
4	Amulya	NULL	NULL
5	Nihitha	NULL	NULL

Query 10: - LEFT JOIN

```sql
SELECT *
FROM class
LEFT JOIN class_info
ON class.id=class_info.id
WHERE class_info.id IS NULL;
```

Output: -

24 14:18:46 SELECT * FROM class LEFT JOIN class_info ON... 2 row(s) returned 0.000 sec / 0.000 sec

id	name	id	address
4	Amulya	NULL	NULL
5	Nihitha	NULL	NULL

Query 11: - RIGHT JOIN

```sql
SELECT *
FROM class
RIGHT OUTER JOIN class_info
ON class.id=class_info.id;
```

Output: -

25 14:20:51 SELECT * FROM class RIGHT OUTER JOIN clas... 5 row(s) returned

id	name	id	address
1	Vidhath		HYDERABAD
2	Raghuveer		DELHI
NULL	NULL	3	CHENNAI
NULL	NULL	7	BENGALURU
NULL	NULL	8	PUNE

Query 12: -UNION

```
2520030597-Vidhath
CREATE TABLE first_table(
id INT,
name VARCHAR(30)
);
CREATE TABLE second_table(
id INT,
name VARCHAR(30)
);
INSERT INTO first_table VALUES
(1,'Akshaya'),
(2,'Manasvi');
INSERT INTO second_table VALUES
(2,'Manasvi'),
(3,'Ramya');
```

28	14:25:46	CREATE TABLE first_table(id INT, name VARCH...	0 row(s) affected	0.047 sec
29	14:25:50	CREATE TABLE second_table(id INT, name VA...	0 row(s) affected	0.031 sec
30	14:25:54	INSERT INTO first_table VALUES (1,'Akshaya'), (...	2 row(s) affected Records: 2 Duplicates: 0 Warni...	0.046 sec
31	14:25:58	INSERT INTO second_table VALUES (2,'Manasv...	2 row(s) affected Records: 2 Duplicates: 0 Warni...	0.047 sec

```
SELECT * FROM first_table
UNION
SELECT * FROM second_table;
```

32	14:27:30	SELECT * FROM first_table UNION SELECT * FR...	3 row(s) returned	0.031 sec / 0.000 sec

Result Grid

id	name
1	Akshaya
2	Manasvi
3	Ramya

Query 13: - UNION ALL

```
2520030597-Vidhath
SELECT * FROM first_table
UNION ALL
SELECT * FROM second_table;
```

Output: -

33 14:29:01 SELECT * FROM first_table UNION ALL SELECT... 4 row(s) returned

id	name
1	Akshaya
2	Manasvi
2	Manasvi
3	Ramya

Query 14: -INTERSECT (MYSQL DO NOT SUPPORT INTERSECT DIRECTLY)

```
SELECT first_table.*
FROM first_table
INNER JOIN second_table
ON first_table.id = second_table.id;
```

Output: -

34 14:32:12 SELECT first_table.* FROM first_table INNER JOI... 1 row(s) returned

id	name
2	Manasvi

Query 15: -MINUS (MYSQL DO NOT SUPPORT MINUS DIRECTLY)

```
SELECT first_table.*
FROM first_table
LEFT JOIN second_table
ON first_table.id = second_table.id
WHERE second_table.id IS NULL;
```

Output: -

35 14:34:55 SELECT first_table.* FROM first_table LEFT JOIN... 1 row(s) returned

id	name
1	Vidhath

Query 16: -Generate a report showing all students and address availability status.

```
2520030597-Vidhath
SELECT c.id,
       c.name,
       CASE
           WHEN ci.address IS NULL
           THEN 'Address Missing'
           ELSE 'Address Available'
       END AS Status
FROM class c
LEFT JOIN class_info ci
ON c.id=ci.id;
```

Output: -

37 14:37:51 SELECT c.id, c.name, CASE WH... 4 row(s) returned 0.000 s

id	name	Status
1	Vidhath	Address Available
2	Raghuveer	Address Available
4	Amulya	Address Missing
5	Nihitha	Address Missing

Query 17: -FULL OUTER JOIN (MYSQL DO NOT DIRECTLY SUPPORT FULL OUTER JOIN)

```
2520030597-Vidhath
xxxxxxxxxx-Ashwini
SELECT *
FROM class
LEFT JOIN class_info
ON class.id = class_info.id
WHERE class_info.id IS NULL
UNION
SELECT *
FROM class
RIGHT JOIN class_info
ON class.id = class_info.id
WHERE class.id IS NULL;
```

Output: -

38	14:41:25	SELECT * FROM class LEFT JOIN class_info ON...	5 row(s) returned	0.000 sec / 0.000 sec

id	name	id	address
4	Amulya	NULL	NULL
5	Nihitha	NULL	NULL
NULL	NULL	3	CHENNAI
NULL	NULL	7	BENGALURU
NULL	NULL	8	PUNE